

Decentralized Lending Platform (Flutter + Firebase)

Overview

A peer-to-peer lending mobile application that enables individuals to lend and borrow money **without relying on traditional banks**.

The platform ensures **trust, transparency, and accountability** through reputation scoring and a public transaction ledger.

Problem Statement

Access to formal banking and credit systems is limited for many individuals. Traditional systems require strict documentation, making it difficult for underserved users to obtain loans.

Solution

This application provides a **decentralized-inspired lending system** where:

- Borrowers can request loans
 - Lenders can fund loans based on trust metrics
 - Transactions are transparently recorded
 - User reliability is maintained through a dynamic credit score
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Key Features

Authentication

- Secure login/signup using Firebase Authentication
- Email/Phone-based authentication

User Identity & Trust

- Profile-based verification
- Dynamic **credit score system** based on user behavior

Lending System

- Borrowers can create loan requests
- Lenders can view and fund loans

Risk Assessment

- Simple rule-based credit scoring:
 - On-time repayment → score increases
 - Late/default → score decreases

Transparent Ledger

- All transactions are recorded in a public ledger
- Ensures visibility and accountability

Repayment Tracking

- Track loan status:
 - Pending
 - Funded
 - Repaid
 - Defaulted
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Tech Stack

Frontend

- Flutter

Backend

- Firebase

Database

- Cloud Firestore

Authentication

- Firebase Authentication

Payments (Optional)

- Razorpay (or simulated transactions)
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User Flow

1. User logs in / signs up
2. Completes profile setup
3. Chooses role (Borrower / Lender)

Borrower:

- Create loan request
- Wait for funding
- Repay loan

Lender:

- View loan requests
 - Evaluate borrower (credit score)
 - Fund loan
 - Track repayments
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Project Structure

lib/

├── screens/

├── models/

├── services/

├── providers/

└── main.dart

Setup Instructions

1. Clone the repository

git clone <repo-url>

2. Install dependencies

flutter pub get

3. Configure Firebase:

- Add google-services.json (Android)
- Add GoogleService-Info.plist (iOS)

4. Run the app

flutter run

Future Enhancements

- Blockchain integration for immutable transactions
 - AI-based credit scoring
 - Smart contract-based lending
 - Real payment gateway integration
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Key Concepts Used

- Peer-to-peer lending
 - Reputation-based trust system
 - Transparent transaction ledger
 - Decentralized finance (DeFi) principles
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Conclusion

This project demonstrates how **technology can enable financial inclusion** by removing dependency on centralized institutions while maintaining trust through transparency and behavioral scoring.

Team

- Add your team member names here
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Note

This is a prototype built for a hackathon and simulates financial transactions for demonstration purposes.